Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: New Project

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 139 kWp

Lagolago - Philippines

Author

Project summary

Geographical Site		**Situation**		**Project settings**	
Lagolago		Latitude	10.75 °N	Albedo	0.20
Philippines		Longitude	124.80 °E		
		Altitude	38 m		
		Time zone	UTC+8		

Weather data
Lagolago
Meteonorm 8.1 (2016-2021), Sat=100% - Synthetic

System summary

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation		**Near Shadings**	**User's needs**
Fixed plane		No Shadings	Unlimited load (grid)
Tilt/Azimuth	15 / 0 °		

System information

PV Array		**Inverters**	
Nb. of modules	231 units	Nb. of units	3 units
Pnom total	139 kWp	Pnom total	135 kWac
		Pnom ratio	1.027

Results summary

Produced Energy	225939 kWh/year	Specific production	1630 kWh/kWp/year	Perf. Ratio PR	80.80 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation

Orientation		**Sheds configuration**	**Models used**	
Fixed plane		No 3D scene defined	Transposition	Perez
Tilt/Azimuth	15 / 0 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

Horizon		**Near Shadings**	**User's needs**
Average Height	3.3 °	No Shadings	Unlimited load (grid)

PV Array Characteristics

PV module		**Inverter**	
Manufacturer	JA solar	Manufacturer	Dasstech
Model	JAM78-S30-600-MR	Model	Soleaf DSP-3345K
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	600 Wp	Unit Nom. Power	45.0 kWac
Number of PV modules	231 units	Number of inverters	3 units
Nominal (STC)	139 kWp	Total power	135 kWac

Array #1 - PV Array			
Number of PV modules	77 units	Number of inverters	1 unit
Nominal (STC)	46.2 kWp	Total power	45.0 kWac
Modules	7 string x 11 In series		
At operating cond. (50°C)		Operating voltage	200-820 V
Pmpp	42.2 kWp	Pnom ratio (DC:AC)	1.03
U mpp	448 V		
I mpp	94 A		

Array #2 - Sub-array #2			
Number of PV modules	77 units	Number of inverters	1 unit
Nominal (STC)	46.2 kWp	Total power	45.0 kWac
Modules	7 string x 11 In series		
At operating cond. (50°C)		Operating voltage	200-820 V
Pmpp	42.2 kWp	Pnom ratio (DC:AC)	1.03
U mpp	448 V		
I mpp	94 A		

Array #3 - Sub-array #3			
Number of PV modules	77 units	Number of inverters	1 unit
Nominal (STC)	46.2 kWp	Total power	45.0 kWac
Modules	7 string x 11 In series		
At operating cond. (50°C)		Operating voltage	200-820 V
Pmpp	42.2 kWp	Pnom ratio (DC:AC)	1.03
U mpp	448 V		
I mpp	94 A		

Total PV power		**Total inverter power**	
Nominal (STC)	139 kWp	Total power	135 kWac
Total	231 modules	Number of inverters	3 units
Module area	646 m²	Pnom ratio	1.03

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Array losses

Thermal Loss factor
Module temperature according to irradiance

Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	79 mΩ
Global wiring resistance	26 mΩ
Loss Fraction	1.5 % at STC

Module Quality Loss

Loss Fraction	-0.8 %

Module mismatch losses

Loss Fraction	2.0 % at MPP

Strings Mismatch loss

Loss Fraction	0.1 %

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

Horizon definition

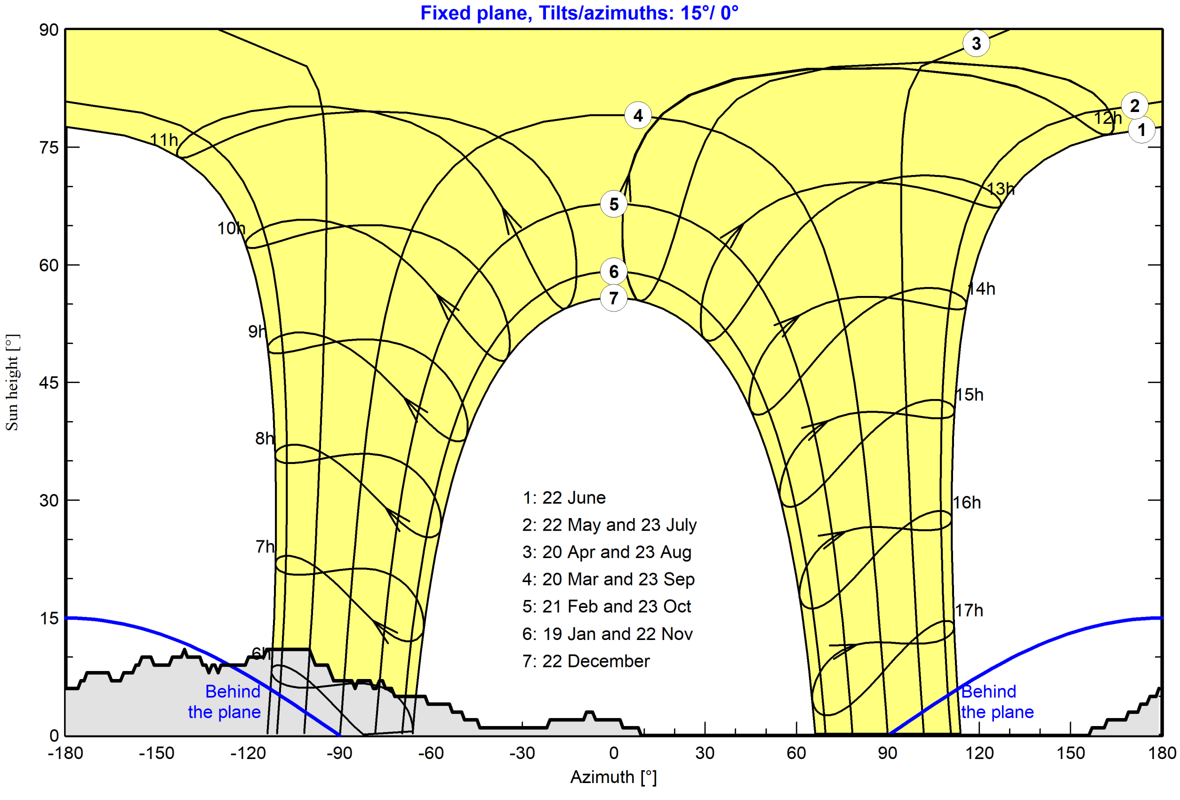
Horizon from Meteonorm web service, lat=10.746, lon=124.7963

Average Height	3.3 °	Albedo Factor	0.94
Diffuse Factor	0.99	Albedo Fraction	100 %

Horizon profile

Azimuth [°]	-180	-176	-174	-173	-162	-161	-158	-156	-155	-152	-146	-145	-142	-140
Height [°]	6.0	6.0	7.0	8.0	7.0	8.0	8.0	9.0	10.0	10.0	9.0	10.0	10.0	10.0
Azimuth [°]	-136	-135	-133	-132	-131	-129	-121	-120	-114	-100	-98	-92	-86	-85
Height [°]	10.0	9.0	8.0	9.0	9.0	9.0	9.0	10.0	11.0	11.0	9.0	7.0	7.0	6.0
Azimuth [°]	-80	-79	-77	-76	-75	-72	-62	-61	-53	-50	-49	-44	-22	-21
Height [°]	7.0	6.0	7.0	7.0	6.0	5.0	5.0	4.0	3.0	3.0	2.0	1.0	1.0	2.0
Azimuth [°]	-9	-7	-6	3	8	9	157	161	162	172	173	176	178	179
Height [°]	3.0	3.0	2.0	1.0	1.0	0.0	1.0	1.0	2.0	3.0	4.0	5.0	5.0	6.0

Sun Paths (Height / Azimuth diagram)



Main results

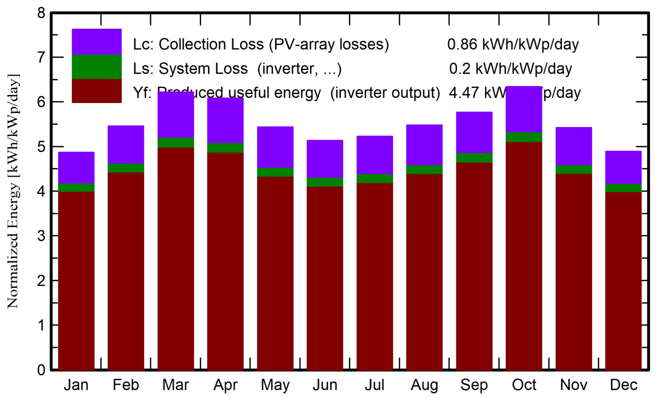
System Production

Produced Energy (P50)	225939 kWh/year	Specific production (P50)	1630 kWh/kWp/year	Perf. Ratio PR	80.80 %
Produced Energy (P90)	201049 kWh/year	Specific production (P90)	1451 kWh/kWp/year		
Produced Energy (P95)	194045 kWh/year	Specific production (P95)	1400 kWh/kWp/year		

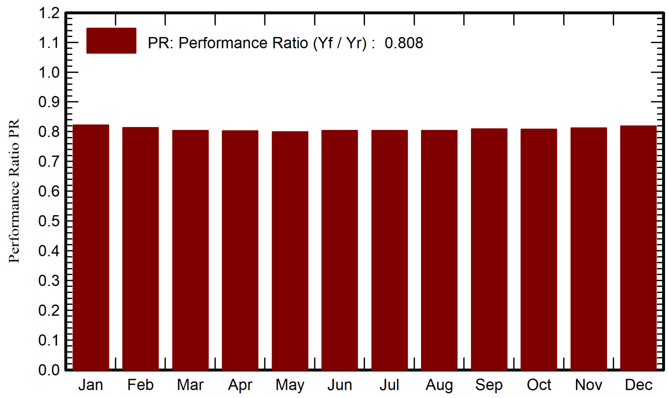
Economic evaluation

Investment		**Yearly cost**		**LCOE**	
Global	2,677,489.00 PHP	Annuities	0.00 PHP/yr	Energy cost	1.18 PHP/kWh
Specific	19.3 PHP/Wp	Run. costs	165,923.44 PHP/yr		
		Payback period	1.6 years		

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	136.7	73.02	25.52	151.1	147.0	18013	17211	0.822
February	142.3	62.97	26.12	152.8	149.0	18016	17226	0.814
March	187.3	73.90	27.27	192.7	187.2	22425	21472	0.804
April	187.8	77.07	28.27	182.6	176.7	21212	20308	0.803
May	181.3	75.43	29.32	168.6	162.7	19540	18663	0.799
June	169.2	74.71	28.49	154.0	148.2	17979	17154	0.804
July	176.8	81.91	28.42	162.1	155.7	18917	18066	0.804
August	178.1	81.01	28.50	169.8	164.0	19792	18917	0.804
September	172.6	79.54	27.47	173.0	167.6	20263	19394	0.809
October	184.4	70.95	27.38	196.6	191.4	22951	22006	0.808
November	146.0	62.35	26.58	162.8	158.0	19151	18323	0.812
December	134.6	70.01	26.14	151.6	146.9	17998	17196	0.819
Year	1997.1	882.87	27.46	2017.5	1954.3	236255	225939	0.808

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

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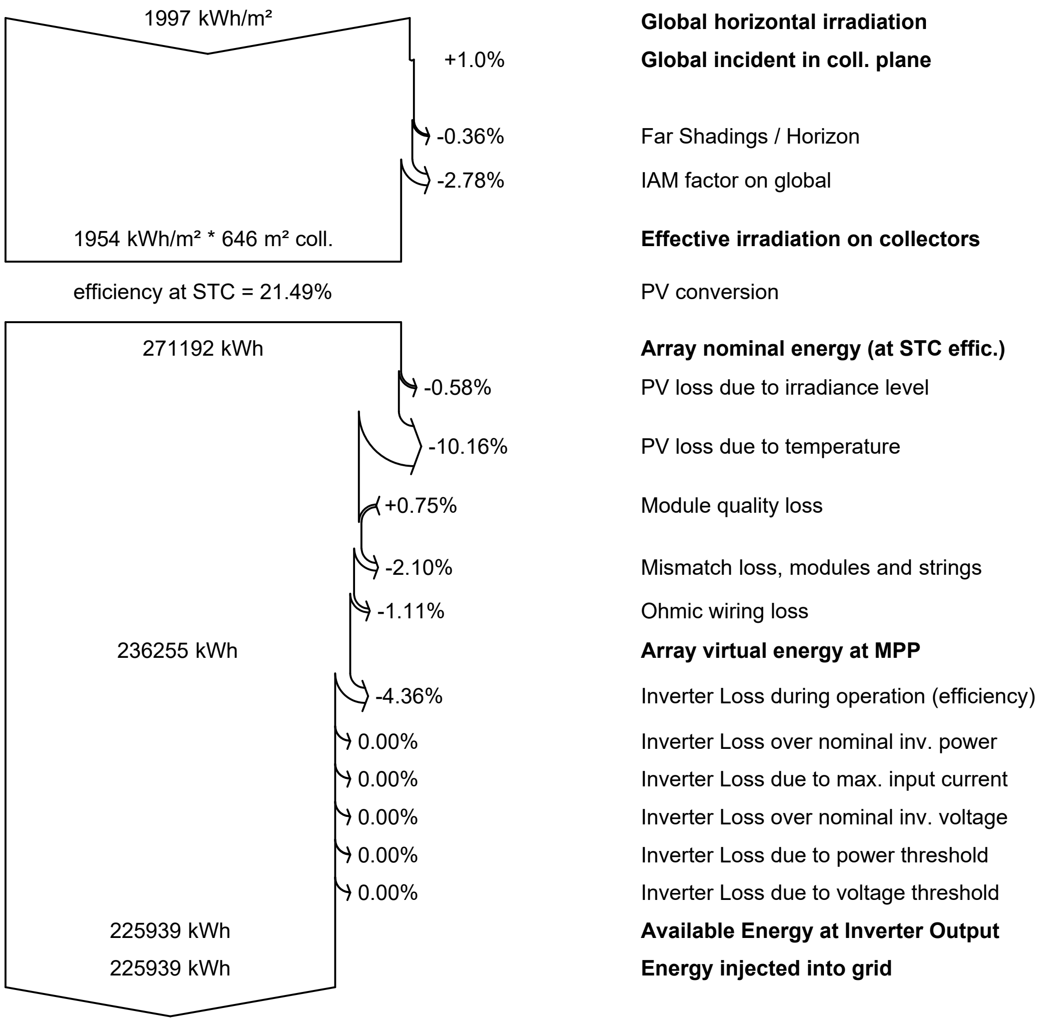
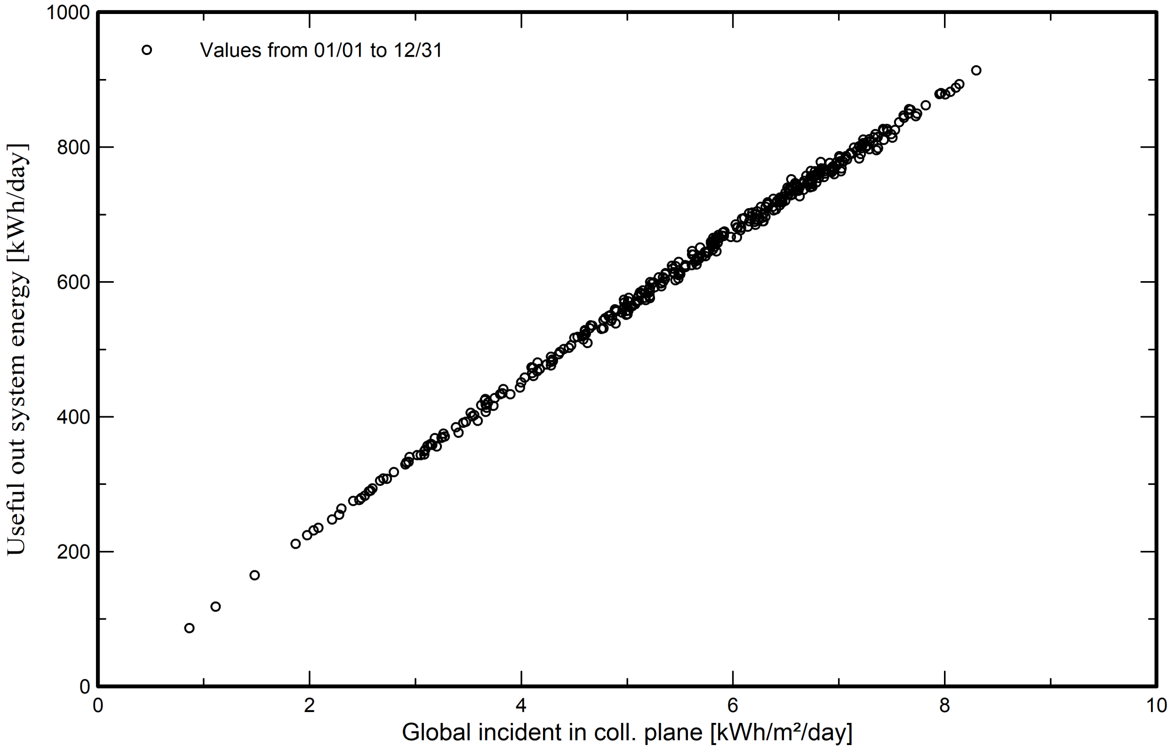
Loss diagram

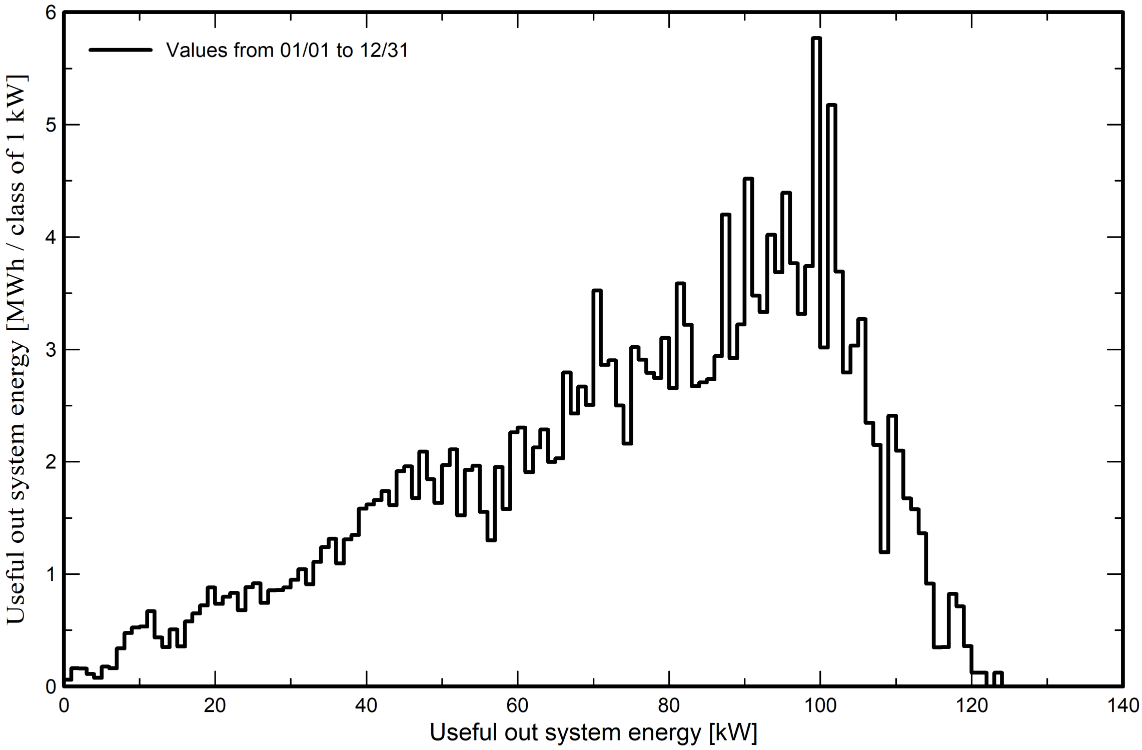


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Meteonorm 8.1 (2016-2021), Sat=100%	
Kind	Monthly averages	
Synthetic - Multi-year average		
Year-to-year variability(Variance)		8.4 %

Specified Deviation

Climate change	0.0 %

Global variability (weather data + system)

Variability (Quadratic sum)	8.6 %

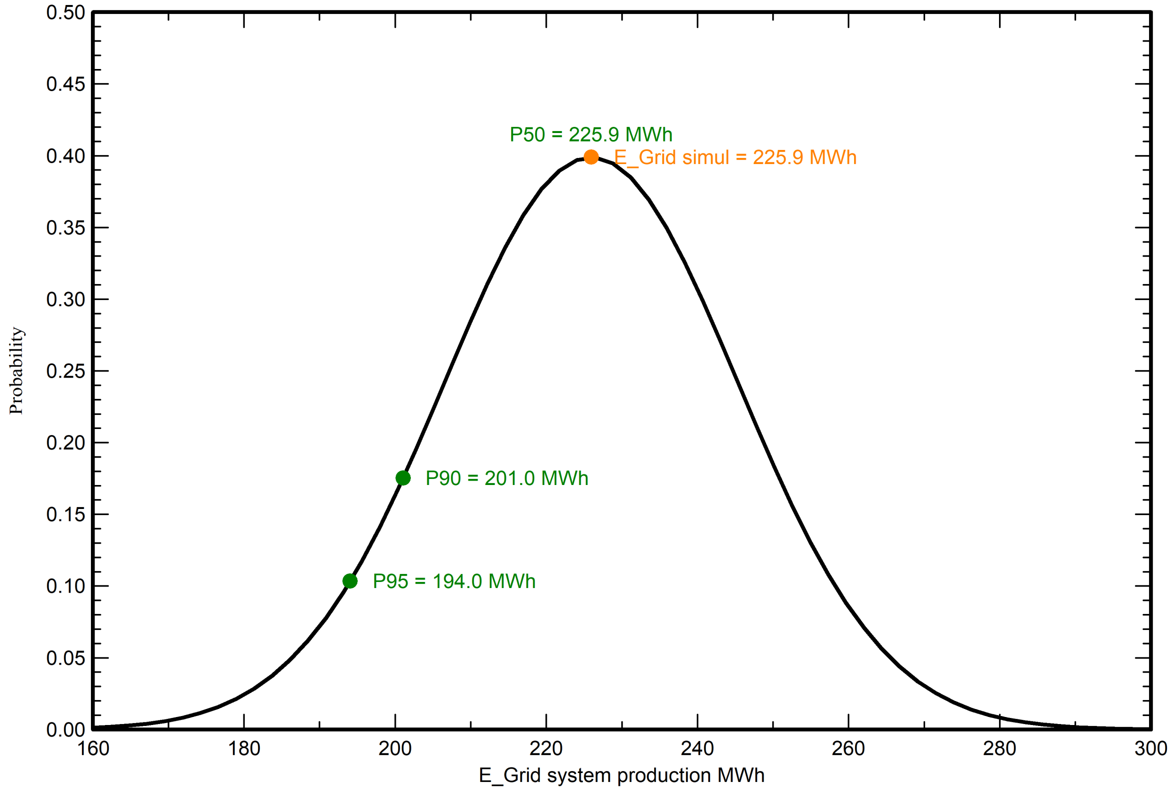
Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

Annual production probability

Variability	19.4 MWh
P50	225.9 MWh
P90	201.0 MWh
P95	194.0 MWh

Probability distribution



Single-line diagram

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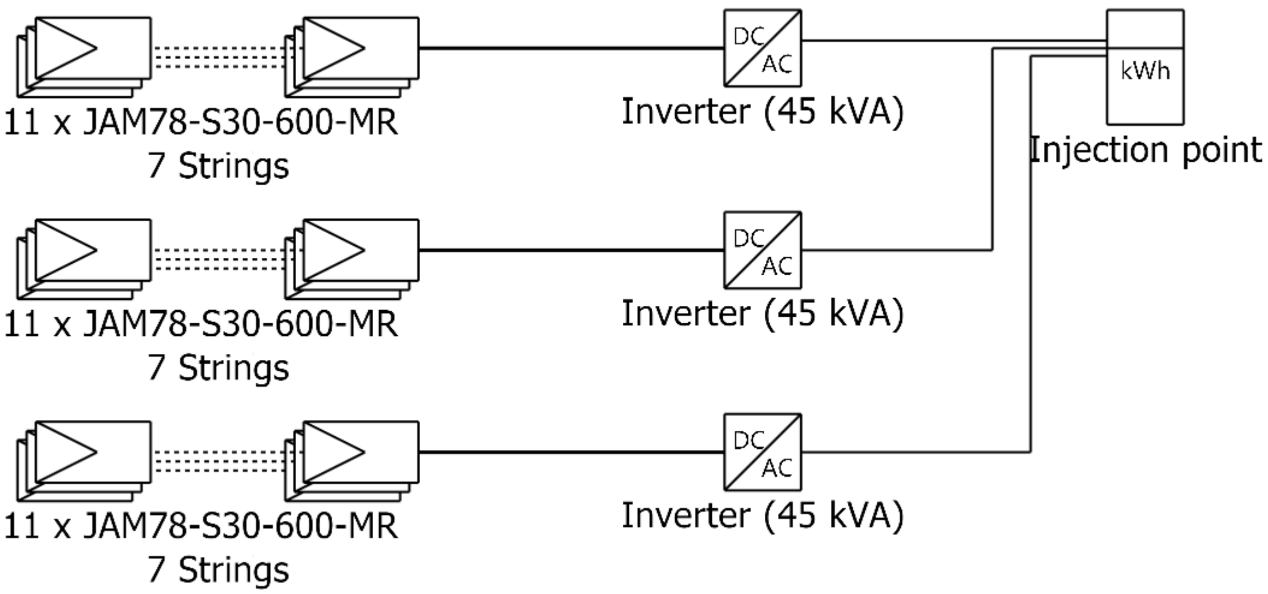

PV module	JAM78-S30-600-MR
Inverter	Soleaf DSP-3345K
String	11 x JAM78-S30-600-MR

New Project	
VC0 : New simulation variant	06/06/26

Cost of the system

Installation costs

Item	Quantity units	Cost PHP	Total PHP
PV modules			
JAM78-S30-600-MR	231	8,319.00	1,921,689.00
Supports for modules	72	900.00	64,800.00
Inverters			
Soleaf DSP-3345K	3	204,500.00	613,500.00
Other components			
Accessories, fasteners	1	20,000.00	20,000.00
Wiring	1	5,000.00	5,000.00
Combiner box	1	30,000.00	30,000.00
Monitoring system, display screen	1	9,000.00	9,000.00
Surge arrester	1	10,000.00	10,000.00
Studies and analysis			
Engineering	1	2,000.00	2,000.00
Permitting and other admin. Fees	1	1,500.00	1,500.00
		Total	2,677,489.00
		Depreciable asset	2,619,989.00

Operating costs

Item	Total PHP/year
Maintenance	
Provision for inverter replacement	122,700.00
Total (OPEX)	122,700.00
Including inflation (2.00%)	165,923.44

System summary

Total installation cost	2,677,489.00 PHP
Operating costs (incl. inflation 2.00%/year)	165,923.44 PHP/year
Produced Energy	226 MWh/year
Cost of produced energy (LCOE)	1.1828 PHP/kWh

Financial analysis

Simulation period

Project lifetime	30 years	Start year	2027

Income variation over time

Inflation	2.00 %/year
Production variation (aging)	0.00 %/year
Discount rate	1.00 %/year

Income dependent expenses

Income tax rate	0.00 %/year
Other income tax	0.00 %/year
Dividends	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (PHP)	Depreciable (PHP)
PV modules				
JAM78-S30-600-MR	Straight-line	20	0.00	1,921,689.00
Supports for modules	Straight-line	20	0.00	64,800.00
Inverters				
Soleaf DSP-3345K	Straight-line	30	0.00	613,500.00
Accessories, fasteners	Straight-line	20	0.00	20,000.00
		Total	0.00	2,619,989.00

Financing

Own funds	2,677,489.00 PHP

Electricity sale

Feed-in tariff	Peak tariff	8.00000 PHP/kWh	
	Off-peak tariff	3.00000 PHP/kWh	18:00-06:00
Duration of tariff warranty		20 years	
Annual connection tax		0.00 PHP/kWh	
Annual tariff variation		0.0 %/year	
Feed-in tariff decrease after warranty		0.00 %	

Return on investment

Payback period	1.6 years
Net present value (NPV)	39,749,930.11 PHP
Internal rate of return (IRR)	62.78 %
Return on investment (ROI)	1484.6 %

Financial analysis

Detailed economic results (kPHP)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	2,677,489	0	0	0	0	0	-2,677,489	0.0%
1	1,807,482	0	122,700	120,774	1,564,007	0	1,684,782	-1,009,388	62.3%
2	1,807,482	0	125,154	120,774	1,561,553	0	1,682,328	639,791	123.9%
3	1,807,482	0	127,657	120,774	1,559,050	0	1,679,825	2,270,212	184.8%
4	1,807,482	0	130,210	120,774	1,556,497	0	1,677,271	3,882,037	245.0%
5	1,807,482	0	132,814	120,774	1,553,893	0	1,674,667	5,475,425	304.5%
6	1,807,482	0	135,471	120,774	1,551,236	0	1,672,011	7,050,535	363.3%
7	1,807,482	0	138,180	120,774	1,548,527	0	1,669,301	8,607,522	421.5%
8	1,807,482	0	140,944	120,774	1,545,763	0	1,666,538	10,146,542	479.0%
9	1,807,482	0	143,763	120,774	1,542,945	0	1,663,719	11,667,747	535.8%
10	1,807,482	0	146,638	120,774	1,540,069	0	1,660,844	13,171,287	591.9%
11	1,807,482	0	149,571	120,774	1,537,137	0	1,657,911	14,657,312	647.4%
12	1,807,482	0	152,562	120,774	1,534,145	0	1,654,920	16,125,969	702.3%
13	1,807,482	0	155,613	120,774	1,531,094	0	1,651,868	17,577,404	756.5%
14	1,807,482	0	158,726	120,774	1,527,982	0	1,648,756	19,011,761	810.1%
15	1,807,482	0	161,900	120,774	1,524,807	0	1,645,582	20,429,181	863.0%
16	1,807,482	0	165,138	120,774	1,521,569	0	1,642,344	21,829,807	915.3%
17	1,807,482	0	168,441	120,774	1,518,266	0	1,639,041	23,213,776	967.0%
18	1,807,482	0	171,810	120,774	1,514,898	0	1,635,672	24,581,226	1018.1%
19	1,807,482	0	175,246	120,774	1,511,461	0	1,632,236	25,932,293	1068.5%
20	1,807,482	0	178,751	120,774	1,507,956	0	1,628,731	27,267,110	1118.4%
21	1,807,482	0	182,326	20,450	1,604,706	0	1,625,156	28,585,811	1167.6%
22	1,807,482	0	185,972	20,450	1,601,059	0	1,621,509	29,888,525	1216.3%
23	1,807,482	0	189,692	20,450	1,597,340	0	1,617,790	31,175,383	1264.4%
24	1,807,482	0	193,486	20,450	1,593,546	0	1,613,996	32,446,511	1311.8%
25	1,807,482	0	197,355	20,450	1,589,676	0	1,610,126	33,702,037	1358.7%
26	1,807,482	0	201,302	20,450	1,585,729	0	1,606,179	34,942,085	1405.0%
27	1,807,482	0	205,328	20,450	1,581,703	0	1,602,153	36,166,777	1450.8%
28	1,807,482	0	209,435	20,450	1,577,597	0	1,598,047	37,376,235	1495.9%
29	1,807,482	0	213,624	20,450	1,573,408	0	1,593,858	38,570,580	1540.6%
30	1,807,482	0	217,896	20,450	1,569,135	0	1,589,585	39,749,930	1584.6%
Total	**54,224,448**	**2,677,489**	**4,977,703**	**2,619,989**	**46,626,756**	**0**	**49,246,745**	**39,749,930**	**1584.6%**

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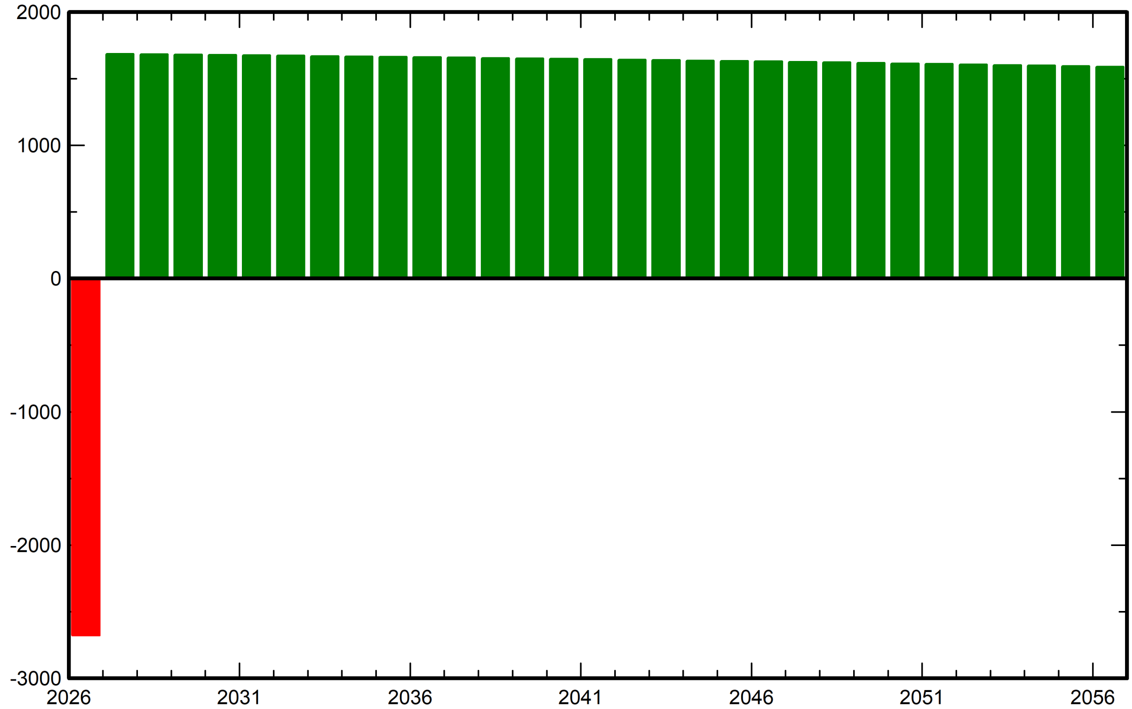
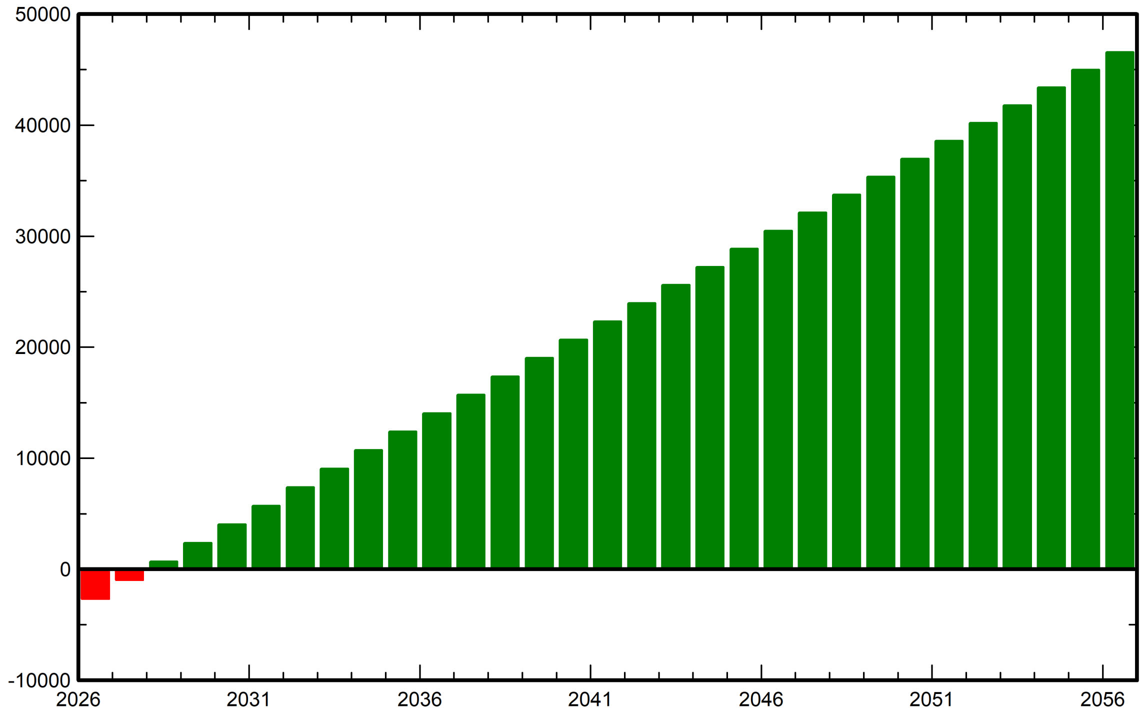
Financial analysis

Yearly net profit (kPHP)



Cumulative cashflow (kPHP)



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CO_2 Emission Balance

Total: 2746.4 tCO_2

Generated emissions

Total: 76.61 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total: 3253.5 tCO_2

System production: 225.94 MWh/yr

Grid Lifecycle Emissions: 480 gCO_2/kWh

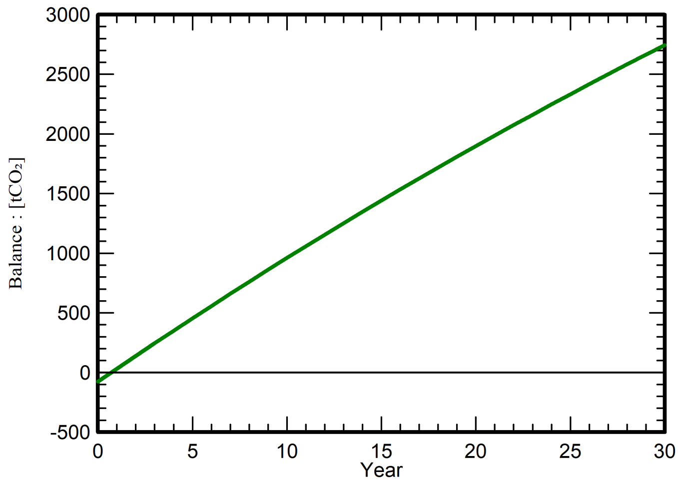
Source: IEA List

Country: Philippines

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	43.2 kWp	73990
Supports	3.20 kgCO2/kg	720 kg	2305
Inverters	317 kgCO2/units	1.00 units	317